

Introduction to Macroeconomics (S102016CR)

Spring Semester

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First Part:
National Accounts
and the Real
Economy

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National Accounts  
and the Real  
Economy"] --> B["Measuring a  
nation's income  
(Chapter 1)"]; A --> C["Measuring the cost  
of living  
(Chapter 2)"]; A --> D["Unemployment  
(Chapter 3)"];
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Measuring a
nation's income
(Chapter 1)

**Measuring the cost
of living
(Chapter 2)**

Unemployment
(Chapter 3)

Week 2:

Measuring the cost of living

In the previous chapter, we saw how to measure « real » income or the « quantity » of goods produced in an economy. The GDP deflator helps us measure the evolution of what was produced and abstract from the effect of price changes on income.

In this chapter, we focus on the evolution of prices, as a measure of the evolution of the cost of living (expenditure rather than production side).

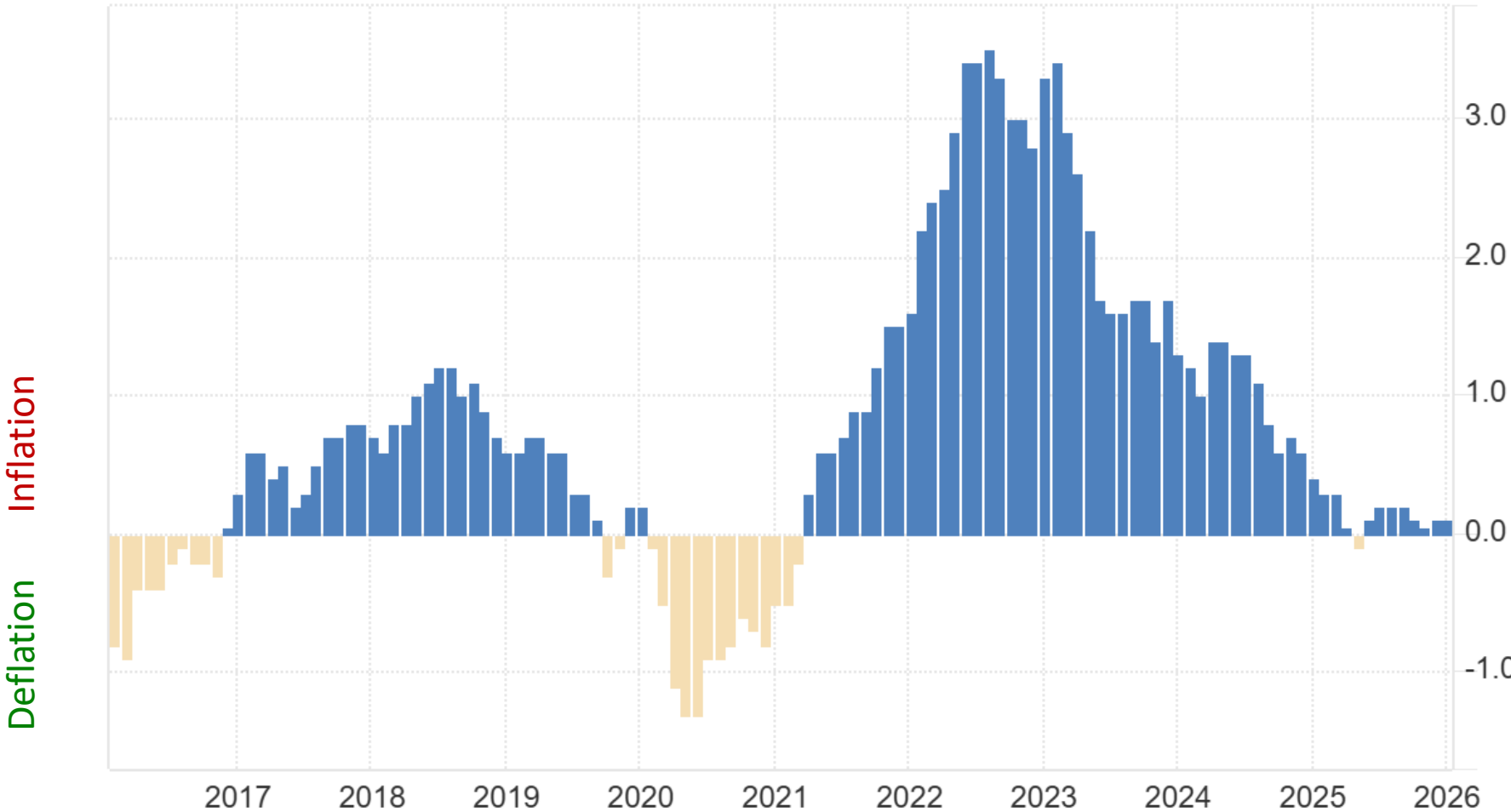
It will be useful to compare the purchasing power of different incomes at different times.

For example, I may prefer an income of 200CHF to an income of CHF2000 if the cost of living when I receive 200CHF is more than ten times smaller than the cost of living when I have an income of CHF2000.

A measure of the evolution of the cost of living are consumer price indices.

They capture the average evolution in consumer prices and therefore « inflation » ($\equiv$ percentage change in the consumer price index between two periods)

Switzerland Inflation Rate (%)



Source: tradingeconomics.com | Swiss Federal Statistical Office

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- a) Consumer Price Indices (CPI)
- b) Problems with CPI
- c) Differences between CPI and GDP deflators
- d) Correcting economic variables for inflation
- e) Summary

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a) Consumer Price Indices (CPI)

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a. Consumer Price Indices

The Consumer Price Index (CPI) captures the general evolution of consumer prices in the economy and therefore it is a good measure of inflation.

It captures the average change in prices from one period to another, and it is therefore a good indicator of the cost of living for the average consumer. If the CPI increases, the average consumer will have to spend more to buy the same consumption basket, and therefore his cost of living will increase.

Crucial inputs to **indexing** prices in the economy:

- wages, pensions
- prices of utilities

But also, to:

- adapt monetary policies (Central Banks) or
- determine investment decisions by entrepreneurs
- determine government policies.

Important to get
it right!

National statistical offices report monthly data on CPI ([FSO in Neuchâtel](#)).

Constructing CPI

1. We first build a representative and fixed consumption basket by surveying consumers and seeing which are the goods and services that enter predominantly into the representative consumer's basket
2. We do regular price surveys to see how the prices of these goods and services are changing
3. We calculate the cost of the representative fixed consumption basket at each period
4. We chose a base year and set the cost of the fixed consumption basket to 100. And we then calculate the value of the index each year by taking the ratio of the cost of the consumption basket each year relative to the base year and multiplying it by 100.

Measuring Inflation

5. The percentage change in the CPI gives the inflation rate:

$$\text{Inflation} = (\text{CPI}_t - \text{CPI}_{t-1}) / \text{CPI}_{t-1} * 100$$

Example

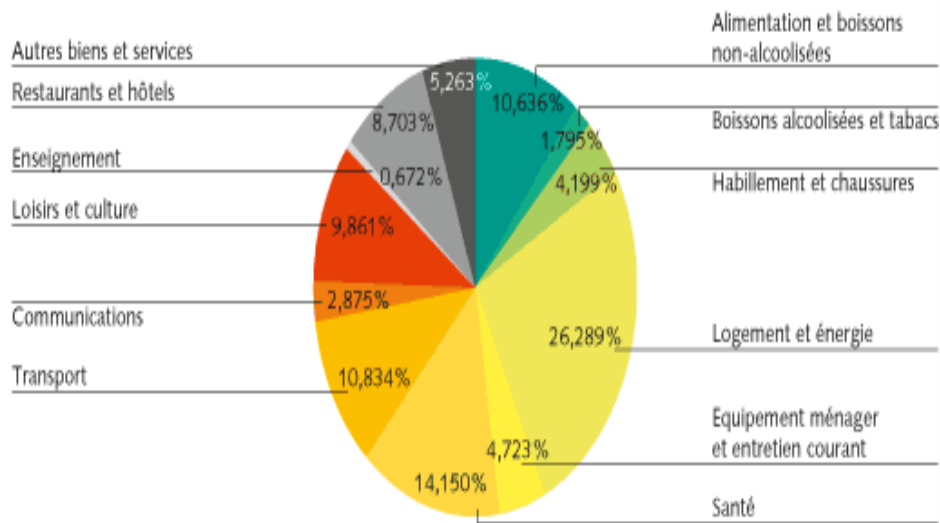
Step 1: Survey Consumers to Determine a Fixed Basket of Goods

4 hot dogs, 2 hamburgers

More generally, they look like this:

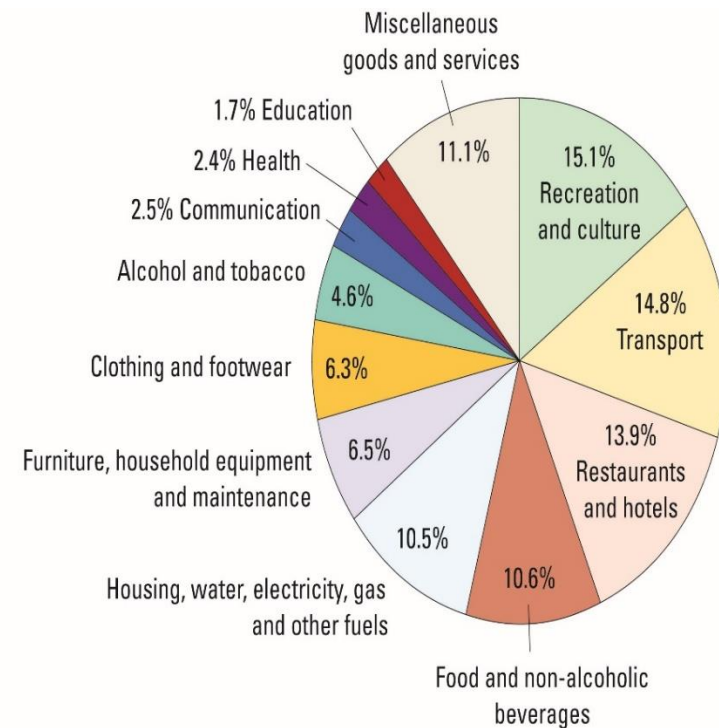
IPC

Structure globale et pondération du panier-type 2011



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Source: FSO, Neuchatel



Source: UK Office for National Statistics.

Step 2: Find the price of each good in each year

Year	Price of hot dogs	Price of hamburgers
2004	€1	€2
2005	2	3
2006	3	4

Step 3: Compute the cost of the basket of goods in each year

2004	$(€1 \text{ per hot dog} \times 4 \text{ hot dogs}) + (€2 \text{ per hamburger} \times 2 \text{ hamburgers}) = €8$
2005	$(€2 \text{ per hot dog} \times 4 \text{ hot dogs}) + (€3 \text{ per hamburger} \times 2 \text{ hamburgers}) = €14$
2006	$(€3 \text{ per hot dog} \times 4 \text{ hot dogs}) + (€4 \text{ per hamburger} \times 2 \text{ hamburgers}) = €20$

Step 4: Choose one year as a base year (2004) and compute the consumer price index in each year

2004	$(€8/€8) \times 100 = 100$
2005	$(€14/€8) \times 100 = 175$
2006	$(€20/€8) \times 100 = 250$

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year

2005	$(175 - 100)/100 \times 100 = 75\%$
2006	$(250 - 175)/175 \times 100 = 43\%$

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b. Problems with CPI

The CPI is an accurate measure of the cost of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.

There are at least four problems or biases with the use of CPI to measure the cost of living:

- Substitution bias
- Introduction of new goods
- Unmeasured quality changes
- Consumer heterogeneity

Substitution bias: The basket does not change to reflect consumer reaction to changes in relative prices.

- Consumers substitute goods that have become relatively less expensive.
- The index overstates the increase in the cost of living by not considering consumer substitution (including “outlet substitution” towards discount shops).

Introduction of New Goods. The basket does not reflect the change in purchasing power brought on by introducing new products.

- New products add greater variety, making each CHF more valuable (movie entrance versus internet viewing).
- Consumers need less money to maintain any given standard of living. The CPI is again over-estimated

Unmeasured Quality Changes. If the quality of good rises (falls) from one year to the next, the value of a CHF rises (falls), even if the price of the good stays the same.

- The FSO tries to adjust prices for consistent quality, but such differences are hard to measure.
- In the IT sector this is a growing problem as changes occur fast.

Consumer heterogeneity: The fixed basket cannot represent everyone (old versus young, rich versus poor, etc.); therefore, the CPI will overestimate changes in the cost of living for some and underestimate it for others.

Response of statistical offices to these problems

- To address substitution bias and new goods, they adjust the consumption basket more often (yearly in Switzerland since 2000). The index becomes a chain index, but the idea is to better capture changes in the cost of living (not the true price change).
- To address changes in quality, adjust the changes in prices by subtracting or adding from the observed price changes, the corresponding change in price due to changes in quality:

$$\Delta P^{\text{adj}} = \Delta P^{\text{obs}} - \Delta P^{\Delta \text{quality}}$$

- Several methods to calculate $\Delta P^{\Delta \text{quality}}$
 - Overlap pricing: difference in prices between old and new products is attributed to quality differences (difficult if both goods are not observed at the same time)
 - Hedonic statistical methods to compute price differences due to different product characteristics (complex if new characteristics appear)

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c. Differences between CPI and GDP deflators

- The GDP deflator is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Two differences with CPI

- First, the *GDP deflator* reflects the prices of all goods and services *produced domestically*, whereas the *consumer price index* reflects the prices of all goods and services *bought by consumers*.
 - Thus, the price of imported goods enters the CPI, but not the GDP deflator,
 - And the price of satellites produced domestically will enter the GDP deflator, but not the CPI
- Second, the *consumer price index* compares the price of a *fixed basket* of goods and services to the price of the basket in the base year (only occasionally does the NSO change the basket), whereas the *GDP deflator* compares the price of *currently produced* goods and services to the price of the same goods and services in the base year.

The CPI in year t is given by:

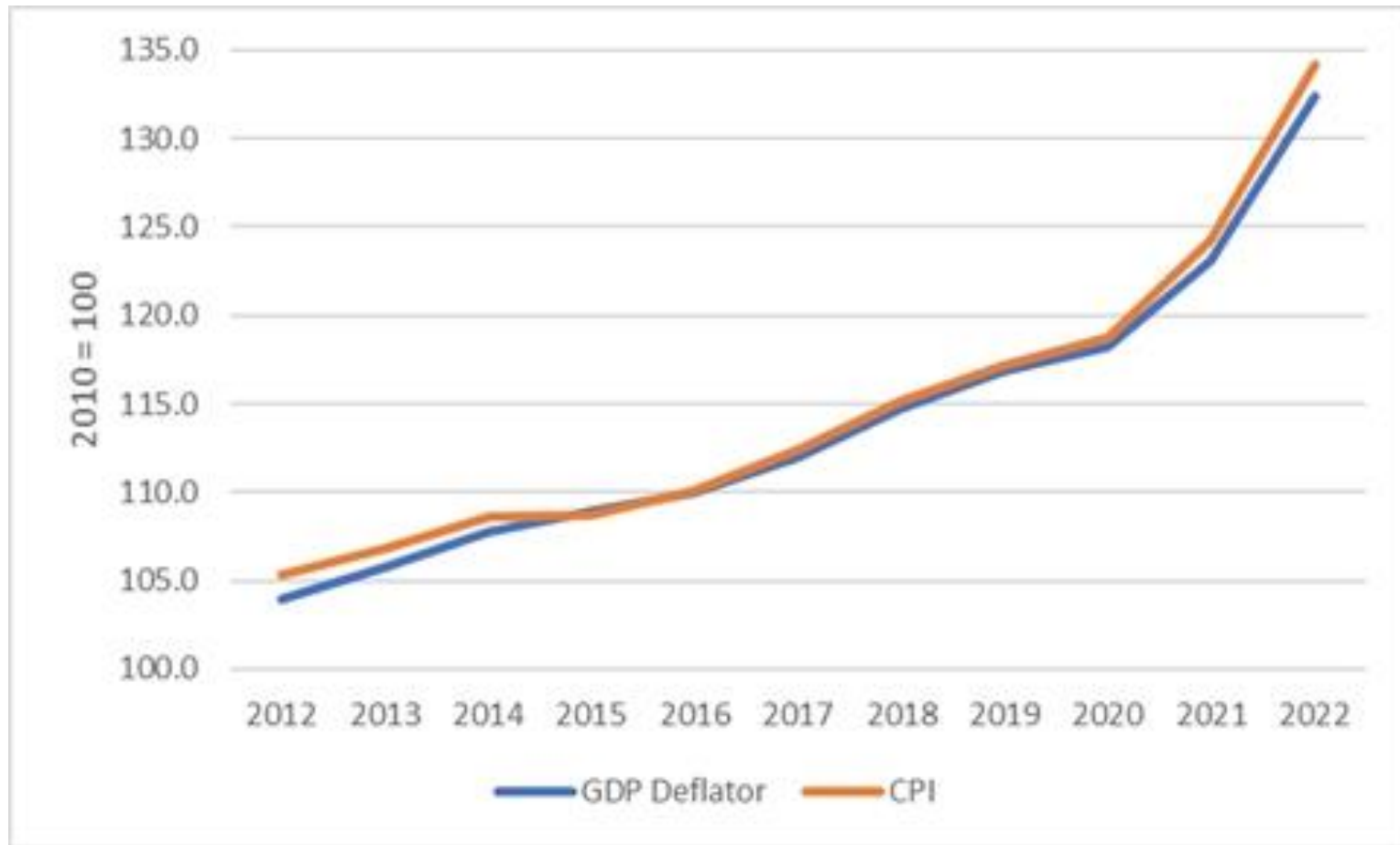
$$\text{CPI}_t = \frac{\text{Cost in } t \text{ of fixed bundle in } 0}{\text{Cost in } 0 \text{ of fixed bundle in } 0} = \frac{\sum_i p_i^t c_i^0}{\sum_i p_i^0 c_i^0}$$

The GDP deflator in year t is given by:

$$\text{GDP Deflator}_t = \frac{\text{Nominal GDP}}{\text{Real GDP}} = \frac{\sum_i p_i^t q_i^t}{\sum_i p_i^0 q_i^t}$$

- The first difference between CPI and the GDP deflator is that the former looks at the evolution of prices in the **consumption** (c) bundle and the second looks at the evolution of prices in the **production** bundle (q).
- The second difference is that the CPI is a **Laspeyres price index** (the bundle is fixed at the initial year), whereas the GDP deflator is a **Paasche price index** (the bundle is fixed at the current year)

CPI versus GDP deflator in the US



- There is at least one-third important Price index: the Producer Price index
- PPI measures the average change over time in the selling prices received by domestic producers for their output.
- The prices included in the PPI are from the first commercial transaction for many products and some services.
- It includes the prices of goods sold exclusively among firms as intermediate inputs.
- Because of this it is thought as a leading indicator of the CPI as producer prices partly capture input costs for producers which later are passed onto consumers.
- But as the GDP deflator it is not representative of the basket of goods purchased by consumers

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d. Correcting economic variables for inflation

To correctly measure the purchasing power of two incomes at two different points in time, we need to correct salaries for inflation:

- The annual salary of George Washington in 1789 was USD 25,000.
- The annual salary of Donald Trump in 2027 is USD 400,000
- CPI with base 100 in 1789 reached 2000 in 2027.

Which of the two had a higher purchasing power, i.e., a higher real income?

$$W \text{ salary in 2027} = \text{Salary in 1789} * (1 + \text{Inflation 2027-1789})$$

$$\begin{aligned} W \text{ salary in 2027} &= \text{Salary in 1789} * (\text{CPI in 2025} / \text{CPI in 1789}) \\ &= 25,000 * 2000 / 100 = 500,000 \end{aligned}$$

$$T \text{ salary in 1789} = \text{Salary in 2027} / (1 + \text{Inflation 2027-1789})$$

$$\begin{aligned} T \text{ salary in 1789} &= \text{Salary in 2027} / (\text{CPI in 2025} / \text{CPI in 1789}) \\ &= 400,000 / 2000 * 100 = 20,000 \end{aligned}$$

In real terms, Washington made 25% more than Trump

Many salaries are contractually **indexed** for inflation (public employees, pensions, etc....)... Therefore, nominal salaries are adjusted for inflation.

The evolution of real and nominal economic variables other than GDP can also differ.

Like with GDP **nominal** variables are measured in current prices, whereas **real** economic variables are measured using the prices of the base year.

$$\text{Nominal wages}_t = \text{Real wages}_t * (1 + \text{Inflation}_t) = \text{Real wages} * (\text{CPI}_t / \text{CPI}_0)$$

which implies that nominal equals real times prices (current CPI with respect to CPI in base year which equals 100).

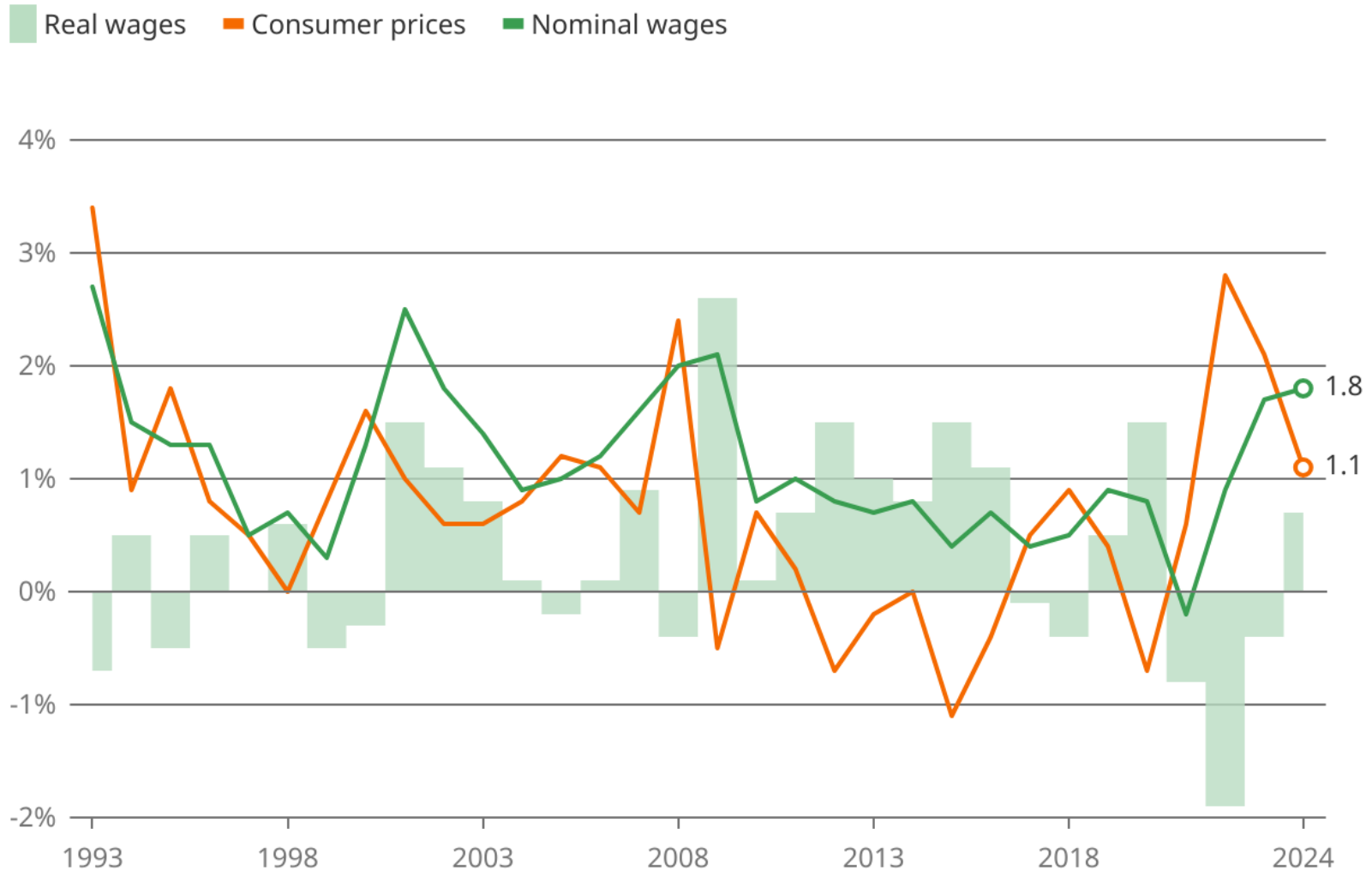
Take logs on both sides: $\ln(\text{nominal}) = \ln(\text{real}) + \ln(\text{CPI}_t / \text{CPI}_0)$

Take the derivative with respect to time (first-order approximation):

$$\% \text{ nominal} = \% \text{ real} + \text{inflation}$$

Evolution of nominal wages, consumer prices and real wages, 1993–2024

Change compared to previous year



Data as on: 22.04.2025

Source: FSO – Swiss Wage Index (SWI)

gr-e-03.04.03.01

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- When making an **investment decision**, what matters is not the nominal returns, but real returns adjusted for inflation
- The *nominal interest* rate is not corrected for inflation (e.g., the interest rate a bank pays).
- The *real interest rate* is the nominal interest corrected for the effects of inflation (also called **Fischer Parity**).

Real interest rate = Nominal interest rates – inflation

Example

- You invest CHF1,000 in a project with a nominal return of 15 percent
- Sounds like a good investment opportunity.
- But if expected inflation is 20%

$$\begin{aligned} \text{Real interest rate} &= \text{Nominal interest rate} - \text{Expected Inflation} \\ &= 15\% - 20\% = -5\% \end{aligned}$$

- A project that look profitable at face value has negative returns once you correct for inflation

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e. Summary

- The consumer price index shows the cost of a basket of goods and services relative to the cost of the same basket in the base year.
- The index is used to measure the overall level of prices in the economy.
- The percentage change in the CPI measures the inflation rate.
- The consumer price index is an imperfect measure of the cost of living for the following four reasons:
 - substitution bias,
 - the introduction of new goods,
 - and unmeasured changes in quality.
 - consumer heterogeneity
- The GDP deflator differs from the CPI because it includes goods and services produced rather than goods and services consumed. In addition, the CPI uses a fixed basket of goods, while the GDP deflator automatically changes the group of goods and services over time as the composition of GDP changes.

- Money figures from different points in time do not represent a valid comparison of purchasing power.
- Various laws and private contracts use price indexes to correct for the effects of inflation.
- The real interest rate equals the nominal interest rate minus the rate of inflation.